

Kumar Satyam

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Education

Indian Institute Of Technology (Indian School Of Mine) Dhanbad

Expected May 2026

Bachelor of Technology in Computer Science and Engineering (CGPA: 8.59/ 10.00)

Dhanbad, Jharkhand

- **Relevant Coursework:** Data Structures and Algorithms (C++), Intro to CS II (C++), Object Oriented Programming (C++), Operating System, Algorithm Design and Analysis, Computer Organisation, Computer Architecture, Theory of Computation

Technical Skills

Languages: C++, Go, Kotlin, TypeScript

Backend Technologies: gRPC, WebSocket, PostgreSQL, Redis, Docker

Frontend Technologies: React, Android SDK (Jetpack Compose)

Developer Tools: Git, GitHub, Docker

Others: MS Power Point, Time Management, Team Collaboration, Problem Solving, Communication Skills, Leadership

Projects

System Design Simulator: A Reality-Grounded Distributed Systems Platform (In Progress, 25% Complete)

- **Tech Stack:** Go, gRPC, WebSocket, PostgreSQL, Redis, Docker, React (TypeScript), Konva.js.
- **Architecture:** Microservices, API Gateway, **ACID-like Isolation**, **Hardware-Adaptive Tick System**.
 - * Innovated an "**ACID-like Isolation**" architecture for natural, emergent resource contention, a breakthrough inspired by DBMS principles.
 - * Architected and completed the initial implementation of the **4 universal base engines (CPU, Memory, Storage, etc.) in Go**, forming the foundation of a unified, hardware-adaptive simulation system.
 - * Developing a simulation model with a **projected 90-92% engine-level accuracy**, grounded in real hardware profiles (Intel, Samsung) and statistical convergence models.
 - * Currently building the component composition system (Databases, Load Balancers) on top of the completed base engines.

NoRegret: A Serverless Mesh Networking Stack (In Progress, 50% Complete)

- **Tech Stack:** C++, Kotlin, JNI, C++ Actor Framework (CAF), Go (CGO), Yggdrasil, Noise Protocol.
- Architecting a high-performance, serverless mesh networking stack and a social productivity app (Pulse) to operate without internet/cellular infrastructure.
 - * Architected a "**Digital Factory**" C++ engine using the Actor model; completed the core supervised actor pipeline for resilient packet processing.
 - * Integrating 5+ battle-tested components, including **NASA-grade FEC** and **Yggdrasil routing**, to establish a production-grade mesh foundation.
 - * Designed an adaptive **Three-Tier Delivery Algorithm** to maximize resilience, currently implementing the high-confidence (Tier 1) delivery logic.

Achievements

- **JEE Advanced 2022:** Secured All India Rank (AIR) **5007** among 150,000 candidates in one of the world's most challenging engineering entrance examinations.
- **Data Structures Algorithms:** Rated **3-Star on CodeChef** and consistently practice problem-solving, having solved over **300+** questions on platforms like LeetCode and Code Studio.
- **Academic Excellence:** Maintained a cumulative GPA of **8.59/10.0**, with a recent semester GPA of **9.04**, demonstrating consistent high performance in rigorous coursework.

Extracurricular Activities

Volunteer, SRIJAN (East India's largest socio-cultural festival) • Sports: Basketball, Chess